

## TASK-1

| Network Threat                              | Threat Description                                                                                                                                                    | Real-World Example in Popular Apps/Services                                                                                                                                                                                                        |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phishing</b>                             | Deceptive communications (emails, messages, fake links) designed to trick users into revealing sensitive data like login credentials or OTPs.                         | <b>Paytm / Bank Fraud:</b> Fraudsters send SMS messages claiming <i>"Your Paytm KYC has expired! Click here to update immediately or account will be blocked."</i> The link leads to a fake Paytm login page to steal credentials and OTPs.        |
| <b>Distributed Denial of Service (DDoS)</b> | Flooding a server or network with massive automated botnet traffic to overwhelm system resources and force service downtime.                                          | <b>Instagram / Server Outages:</b> Attackers direct millions of simultaneous junk requests to Instagram's API edge servers, causing the app to fail to load feeds, post photos, or send DMs for millions of users worldwide.                       |
| <b>Man-in-the-Middle (MitM)</b>             | An attacker secretly intercepts and potentially alters communication packets between two parties who believe they are directly communicating.                         | <b>Public Wi-Fi &amp; WhatsApp Web:</b> Connect to an rogue/unsecured public Wi-Fi hotspot where an attacker intercepts unencrypted traffic or redirects your browser to a fake WhatsApp Web login screen to capture your session token.           |
| <b>Malware (Spyware / Trojan)</b>           | Malicious software installed on a device without user consent to steal data, track activity, or take control of the system.                                           | <b>Android APKs Targeting Paytm:</b> A user downloads a modified/modded app or fake screen-sharing app (like AnyDesk) via a rogue link. The malware secretly captures the device screen to read incoming Paytm transaction PINs and bank SMS OTPs. |
| <b>Password &amp; Brute Force Attacks</b>   | Automated tools systematically guessing thousands of username-password combinations or using leaked database dumps (credential stuffing) to gain unauthorized access. | <b>Instagram Account Takeovers:</b> Botnets try thousands of leaked passwords per minute against Instagram's login API endpoint to break into accounts that reuse passwords across multiple websites.                                              |

## TASK-2

```
main.py  [ ] [ ] [ ] Share Run Output
1 # Simple Brute-Force PIN Guessing Simulation
2
3 # Preset correct PIN
4 correct_pin = "4827"
5
6 attempts = 0
7
8 # Try every possible 4-digit PIN (0000 to 9999)
9 for pin in range(10000):
10     attempts += 1
11
12     # Format the number as a 4-digit string
13     guess = f"{pin:04d}"
14
15     print(f"Attempt {attempts}: Trying PIN {guess}")
16
17     # Check if the guessed PIN matches the correct PIN
18     if guess == correct_pin:
19         print("\nPIN Found!")
20         print("Correct PIN:", guess)
```

Attempt 4815: Trying PIN 4814  
Attempt 4816: Trying PIN 4815  
Attempt 4817: Trying PIN 4816  
Attempt 4818: Trying PIN 4817  
Attempt 4819: Trying PIN 4818  
Attempt 4820: Trying PIN 4819  
Attempt 4821: Trying PIN 4820  
Attempt 4822: Trying PIN 4821  
Attempt 4823: Trying PIN 4822  
Attempt 4824: Trying PIN 4823  
Attempt 4825: Trying PIN 4824  
Attempt 4826: Trying PIN 4825  
Attempt 4827: Trying PIN 4826  
Attempt 4828: Trying PIN 4827

PIN Found!  
Correct PIN: 4827  
Total Attempts: 4828

=== Code Execution Successful ===

## TASK-3

Physical security is just as critical as cybersecurity when protecting high-availability infrastructure like Zomato or BookMyShow. If an attacker or a physical hazard reaches the physical hardware, software defenses can be completely bypassed.

Here are **4 physical security risks** for a server room and their corresponding mitigation steps:

### 1. Unauthorized Physical Access & Break-Ins

- **The Risk:** An unauthorized individual ( intruder, disgruntled employee, or malicious actor posing as a technician) gains physical entry to the server room. They could tamper with server racks, insert malicious USB drives, unplug network cables, or steal storage drives containing user and payment data.
- **Mitigation:** Implement **Multi-Factor Physical Access Control (Biometrics + Smart Cards)**. Require a combination of keycards and biometric scanning (fingerprint/retina) for entry, keep server racks locked, and maintain automated electronic access logs for every door opening.

### 2. Environmental & Fire Hazards

- **The Risk:** Server racks generate intense heat, and an electrical short-circuit or cooling failure can trigger a fire. Standard water sprinkler systems would destroy millions of rupees worth of active server hardware and cause catastrophic service downtime.

- **Mitigation:** Install **Clean Agent Fire Suppression Systems** (such as FM-200 or Novec 1230 gas systems) paired with **VESDA (Very Early Smoke Detection Apparatus)**. These systems extinguish fires chemically without using water or damaging electronic components.

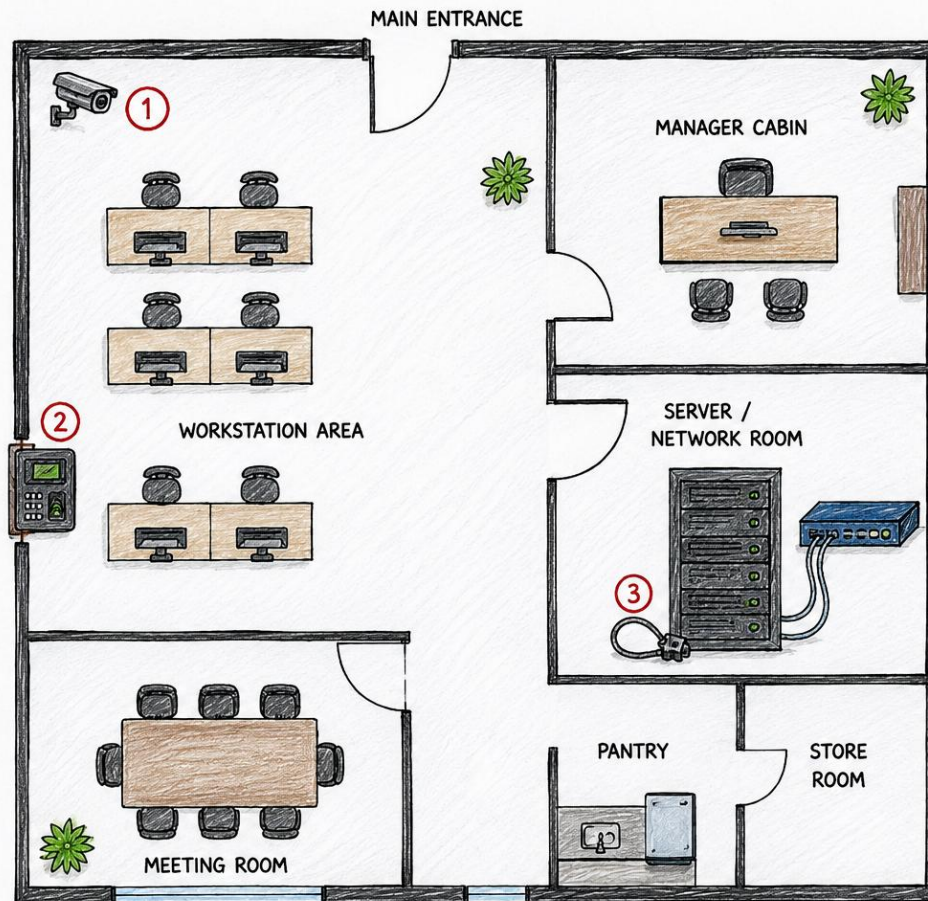
### 3. Power Disruptions & Voltage Spikes

- **The Risk:** Sudden blackouts, power surges, or utility grid failures can cause abrupt server shutdowns. This can lead to database corruption, lost transactions during peak booking/ordering hours, and hardware component failure.
- **Mitigation:** Deploy **Redundant Uninterruptible Power Supplies (UPS)** paired with **Automatic Diesel Generators (N+1 Configuration)**. The UPS provides instantaneous backup battery power to bridge the gap until diesel generators automatically fire up within seconds.

### 4. Cable Tampering & Physical Snooping

- **The Risk:** Exposed or unmonitored fiber/Ethernet cables running along walls or ceilings can be physically spliced, tapped, or cut, allowing an attacker to intercept raw data packets (Man-in-the-Middle attack) or disrupt network connectivity.
- **Mitigation:** Enclose all network cables in **Secured Metal Conduits** and run them under raised floors or inside sealed cable trays. Combine this with continuous 24/7 **CCTV Video Surveillance** with zero blind spots inside and outside the server room.

TASK-4



## PHYSICAL SECURITY MEASURES



### CCTV SURVEILLANCE

CCTV cameras at key locations to monitor and record all activities.



### BIOMETRIC ACCESS

Biometric access control at the main entrance to allow only authorized personnel.



### CABLE LOCKS

Cable locks on network equipment to prevent theft or unauthorized removal.

**Note:** Keep server room locked and access restricted to IT personnel only.

## NETWORK & PHYSICAL SECURITY CHECKLIST FOR COLLEGE COMPUTER LAB

| No. |                                                                                    | ACTION                                                                                                                  | SPECIFIC FOR COLLEGE COMPUTER LAB                                                                                                                                                                                                                             |
|-----|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   |   | <b>Keep Systems &amp; Software Updated</b><br>Regularly update OS, applications, and firmware to patch vulnerabilities. | <ul style="list-style-type: none"> <li>• Enable automatic updates for Windows and lab software.</li> <li>• Update antivirus and firewall definitions regularly on all systems.</li> </ul>                                                                     |
| 2   |   | <b>Configure &amp; Monitor Firewalls</b><br>Use firewalls and monitor traffic to block unauthorized access.             | <ul style="list-style-type: none"> <li>• Enable Windows Firewall on all PCs and the server.</li> <li>• Restrict unnecessary ports; allow only required services (e.g., DNS, DHCP).</li> <li>• Review firewall logs weekly for suspicious activity.</li> </ul> |
| 3   |   | <b>Use Strong Access Control</b><br>Limit access to network resources and administrative accounts.                      | <ul style="list-style-type: none"> <li>• Use strong passwords and change them periodically.</li> <li>• Create separate accounts for students and staff.</li> <li>• Enable screen lock with auto-timeout on all lab PCs.</li> </ul>                            |
| 4   |   | <b>Segment the Network</b><br>Separate student devices from admin systems and sensitive data.                           | <ul style="list-style-type: none"> <li>• Use VLANs: one for student PCs, one for staff, one for servers.</li> <li>• Block communication between student VLAN and admin VLAN.</li> </ul>                                                                       |
| 5   |   | <b>Secure Physical Access</b><br>Prevent unauthorized entry to the lab and server/network equipment.                    | <ul style="list-style-type: none"> <li>• Keep the lab door locked; use access card/biometric if available.</li> <li>• Allow entry only to authorized students and staff.</li> <li>• Keep visitors under staff supervision.</li> </ul>                         |
| 6   |   | <b>Protect Network Equipment</b><br>Secure routers, switches, and cables from tampering or theft.                       | <ul style="list-style-type: none"> <li>• Place network devices in a locked cabinet or rack.</li> <li>• Use cable locks for switches, routers, and access points.</li> <li>• Keep spare ports disabled and check cables regularly.</li> </ul>                  |
| 7   |  | <b>Backup, Monitor &amp; Educate</b><br>Back up important data, monitor systems, and promote security awareness.        | <ul style="list-style-type: none"> <li>• Backup important data regularly to an external drive or cloud.</li> <li>• Monitor server and network health (logs, alerts).</li> <li>• Educate students about safe internet use and reporting issues.</li> </ul>     |